



## Asset Performance: Ene 17 External lighting



Credits



No Minimum Standard

### Aim

To minimise operational energy consumption and the associated carbon emissions from energy efficiency of external lighting.

### Question

What types of external lighting and car park lighting are installed?

| Credits | Answer | If B is not selected, please select either answer C or D and either answer E or F.                                           |
|---------|--------|------------------------------------------------------------------------------------------------------------------------------|
| 0       | A.     | Question not answered                                                                                                        |
| 0       | B.     | All car park lighting and external lighting is not energy efficient and is not fitted with automatic energy savings controls |
| 1       | C.     | External lighting is present and energy efficient OR is fitted with automatic energy saving controls                         |
| 2       | D.     | External lighting is present and energy efficient AND is fitted with automatic energy saving controls                        |
| 1       | E.     | Car park lighting is present and energy efficient OR is fitted with automatic energy saving controls                         |
| 2       | F.     | Car park lighting is present and energy efficient AND is fitted with automatic energy saving controls                        |

### Assessment criteria

| Criterion | Assessment criteria                                                                                                                                                         | Applicable Answer |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1.        | <b>Filtering</b><br>Where external lighting or car park lighting are not necessary from a safety perspective, the associated credits can be filtered out of the assessment. | All               |
| 2.        | This issue only applies to car parking areas associated with the asset being assessed.                                                                                      | B, E, F           |

| Criterion | Assessment criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Applicable Answer |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 3.        | <p>External lighting and car park lighting are energy efficient where;</p> <p>The average initial luminous efficiency is not less than 70 luminaire lumens per circuit Watt.</p> <p>OR</p> <p>All lighting is provided by the following lamp types:</p> <ul style="list-style-type: none"> <li>• LED</li> <li>• Metal halide</li> <li>• Other lamp types with luminous efficiency is not less than 70 luminaire lumens per circuit Watt.</li> </ul>                                                                                                                                                                                                                                                             | C – F             |
| 4.        | <p>Automatic energy saving lighting controls include automatic on and off control to prevent operation during daylight hours or presence detection in areas of intermittent pedestrian traffic. Presence detectors must be compatible with the lamp type used as very frequent switching can reduce the life of some lamp types. Other forms of presence-related control can be used, provided that they switch off the lighting when nobody is in the space. For external lighting not fitted with presence detectors, time switches must provide automatic switch off lighting after a specified curfew hour, except in cases where there is a specific requirement for lighting to be left on all night.</p> | C – F             |
| 5.        | <p>Car park lighting control is classified as energy-efficient if it is equipped with a timer, motion sensor or dimming control (if applicable).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | E, F              |

## Methodology

### Temporary lighting, decorative lighting and floodlighting:

Decorative lighting and floodlighting must be included in this issue, but temporary lighting such as theatrical stage or local display installations can be excluded.

### Emergency lighting:

Emergency light fittings, including security lighting, that are also used for normal operation must be included in this issue, but lighting which only activates in an emergency can be excluded.

### Car park lighting:

Car park lights on open terrain, open air, and covered car parks must be considered in this issue.

### Average initial luminous efficacy of the external light fittings:

The individual luminous fluxes of all luminaires within the construction zone are summed (in lumens), then divided by the total circuit Watts for all the luminaires.

For lamps other than LED lamps, the luminous flux of a luminaire using those lamps can be determined by multiplying the sum of the luminous fluxes produced by all the lamps in the luminaire by the light output ratio of the luminaire (as confirmed by the luminaire manufacturer).

LED lamps are typically integral to the luminaire (LED luminaires). As such, the manufacturers' literature will encompass both lamp and luminaire as a whole.

## Evidence

| Criteria | Evidence requirement                                                                                                                                                                                             |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -        | The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance. |
| All      | A list of the type of luminaires, the control scheme by the building manager (for example based on the long-term maintenance, operating and maintenance manual or instructions for use of the asset).            |
| All      | Visual inspection and verification through photographic evidence.                                                                                                                                                |

## Definitions

### Areas of intermittent pedestrian traffic:

An area can be considered to have intermittent pedestrian traffic where a pedestrian is in or approaching the space less than two-thirds of the time during the period when the lighting, without presence detection, would be switched on.

### Automatic control:

An automatic external lighting control system prevents operation during daylight hours through either a time switch or a daylight sensor (a manually switched lighting circuit with daylight sensor or time switch override is also acceptable).

### Car park associated with the asset:

A car park is considered to be associated with the asset where it is under the same management as the asset and is intended for the use of asset occupants.

### Daylight sensor:

A type of sensor that detects daylight and switches lighting on at dusk and off at dawn.

### External lighting:

Building lighting and advertising lighting, lighting from the entrance, canopy lighting, lighting of paths, roads, parking lots, garages and other outdoor areas that belong to the plot of the building.

### Luminous efficacy in luminaire lumens per circuit Watt:

The ratio between the luminous flux produced by an entire luminaire (light fitting) (in lumens) and the total power consumed by the lamps and the control gear contained within the luminaire (Watts).

### Presence detector:

A sensor that can turn lighting on when a presence is detected in the scanned area, and off after a pre-set time when no presence is detected. Examples could include absence detection, where the light is switched on using a push button or similar control but switching off is done automatically; and key control in secure areas, where a swipe card or key pad is used on entering a space, and the lighting then comes on and remains on until the space is not occupied.

**Time switch:**

A switch with an inbuilt clock which will allow lighting to be switched on and off at programmed times.